Malayalam Speech AI Research Internship

Experimental Report

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Submission status: This report documents the experiments completed and explicitly identifies assessment requirements that remain incomplete. It does not claim that the 100-question end-to-end QA evaluation was completed.

Abstract

This project investigates a reproducible Malayalam speech-to-text component and a small text-response experiment as a first step toward an Indic speech assistant. On a fixed sample of 100 utterances from the Common Voice Malayalam test split, sajilck/whisper-small-malayalam-v2 achieved WER 23.40% and CER 3.75%, compared with 58.20% WER and 12.67% CER for the baseline. A separate 10-question text-only pilot compared Qwen2.5-1.5B-Instruct without context and with a supplied verified fact. Exact-answer substring match was 0% and 60%, respectively. The pilot is too small and the metric too crude to establish semantic correctness. The 100-question QA evaluation and full end-to-end audio QA pipeline remain incomplete.

1. Objectives

Compare recent speech architectures; evaluate Malayalam ASR on a fixed sample; explore context-assisted factual QA; and document limitations and a practical improvement.

2. Task 1 — Speech LLM architecture comparison

2.1 Moshi (Défossez et al., 2024)

Moshi is a speech-text foundation model designed for full-duplex dialogue. Rather than chaining voice activity detection, ASR, a text dialogue model, and TTS, it jointly models user and assistant speech streams. It uses a text LLM backbone augmented with an audio language model that predicts discrete units from a neural audio codec. A hierarchical semantic/acoustic representation and time-aligned text tokens (“Inner Monologue”) help preserve linguistic content while producing speech. Parallel user/assistant streams enable modeling overlap, interruptions, and interjections. The authors report 160 ms theoretical latency and about 200 ms in practice for their configuration; this is not a guarantee for every deployment. Multi-stream autoregressive prediction is powerful but complex, and language coverage, codec quality, endpointing, and compute remain risks.

2.2 Qwen2.5-Omni (Xu et al., 2025)

Qwen2.5-Omni handles text, audio, images, and video and can generate text and speech in a streaming manner. Its Thinker-Talker architecture assigns multimodal understanding and text generation to the Thinker (a Transformer decoder with audio/visual encoders) and streaming discrete speech-token generation to a dual-track autoregressive Talker. Block-wise encoders support streaming input; TMRoPE aligns audio/video timing; a sliding-window diffusion Transformer decoder reduces initial audio packet delay. This broad multimodal design is flexible but can increase model and deployment complexity. Autoregressive speech token generation, chunk sizes, buffering, and language coverage still affect real-world latency and quality.

2.3 Comparison

Dimension	Moshi	Qwen2.5-Omni
Primary focus	Full-duplex spoken dialogue	Unified audio/text/vision/video and speech output
Representation	Discrete neural codec audio units; parallel user/assistant streams; aligned text	Audio/visual encoder representations; Thinker hidden states; discrete speech tokens
Semantic/acoustic relation	Text inner monologue provides linguistic scaffold; codec units realize acoustic output	Thinker creates semantic/text representation; Talker generates acoustic tokens
Alignment/streaming	Parallel streams support overlap and interruption; streaming recognition/TTS	Block-wise input; TMRoPE for audio-video time alignment; streaming Talker
Generation/stopping	Autoregressive multi-stream tokens; EOS and application turn/interruption controls	Thinker text + autoregressive Talker; streaming/EOS controls and application cancellation
Trade-off	Conversation immediacy vs synchronized-stream complexity	Broad multimodality vs compute and component complexity

2.4 Low-latency conversation: limitations and proposed improvement

Moshi is explicitly aligned with full-duplex conversation and overlap; Qwen2.5-Omni offers a flexible multimodal Thinker-Talker split. Neither paper guarantees low latency or robust Malayalam dialogue in a given deployment. End-to-end latency includes capture, chunking, inference, decoding, transport, and playback. Low-resource language coverage, accent/code-switching, noise, endpointing, barge-in, and hallucinations must be evaluated. Proposed improvement: a streaming, confidence-aware hybrid system that emits partial ASR hypotheses, retrieves trusted facts for factual questions, asks for clarification when transcript/retrieval confidence is low, generates short response chunks, and implements explicit timeout/cancellation/turn-taking policies. Evaluate clean/noisy speech, speaking rates, speakers, and code-switching.

3. Dataset and setup

The ASR data was Mozilla Common Voice Scripted Speech 27.0 Malayalam. The inspected test.tsv contained 894 rows and 13 columns; all 894 referenced audio files existed. A fixed 100-example sample was selected with random seed 42 and reused across model comparisons. The dataset is read/scripted speech, not spontaneous dialogue. The work was run in Google Colab with a Tesla T4 GPU, PyTorch 2.11.0+cu130, and Transformers 5.20.1.

4. ASR results

Model	N	WER	CER	Mean time	Overall RTF
sajilck/whisper-small-malayalam	100	58.20%	12.67%	2.180 s	0.481
sajilck/whisper-small-malayalam-v2	100	23.40%	3.75%	2.208 s	0.486
Whisper baseline, beam=5	100	58.20%	12.672%	2.128 s	0.469

The v2 model reduced WER by approximately 59.8% relative and CER by approximately 70.4% relative on the fixed sample. It did not demonstrate a speed improvement; mean latency and RTF were similar. The beam=5 run produced no changed transcripts among 100 and did not improve observed WER/CER. This is a negative result in this run; the generation configuration may not have applied as intended, so it is not proof that beam search cannot help.

Observed errors included spacing/word-boundary differences, spelling variants, proper names, and acronym confusion. Keep normalization consistent and report raw outputs. Malayalam word segmentation can affect WER, so CER provides a complementary character-level metric.

5. QA pilot

A separate 10-question text-only pilot used Qwen2.5-1.5B-Instruct with deterministic decoding and max_new_tokens=80. The baseline was asked questions without facts; the context-assisted condition received a short verified fact.

Condition	N	Exact-answer substring match	Mean latency
No context	10	0%	2.833 s
Verified context	10	60%	2.131 s

The string-match metric misses valid paraphrases and may count a substring in an otherwise flawed sentence. Manual inspection found incomplete or malformed responses in the context condition. These results support only a cautious pilot observation that context improved exact-string matches. They do not establish semantic correctness or a hallucination-rate reduction. Latency is text-generation-only, not end-to-end speech latency.

6. Limitations and incomplete requirements

The 100-example ASR evaluation is completed, but QA has only 10 text-only examples. The full audio → ASR → LLM end-to-end evaluation, 100-question QA test, two-condition acoustic/linguistic evaluation, and formal hallucination annotation remain incomplete. Common Voice is scripted speech and does not establish spontaneous conversation performance. No fine-tuning was performed. These gaps should be disclosed rather than described as completed.

7. Conclusion

The strongest completed result is the ASR comparison: the v2 Malayalam Whisper checkpoint substantially reduced WER and CER on a fixed sample with similar latency. The QA pilot suggests supplied context can help a small model answer some factual questions, but more examples and human-validated metrics are necessary. The recommended next experiment is a 100-question retrieval-assisted evaluation connected to audio input.

References

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